

Brain Tumor Segmentation and Localization Using Traditional Image Processing and OpenCV

Weekly Status Report Week 4 of 10

Course: ECE501 – Digital Image Processing — Instructor: Professor Mehul Raval — Section: 01

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1 Introduction

This week focused on studying and implementing the segmentation pipelines described in three research papers supplied for the project. The objective was not only to reproduce the individual processing stages, but also to compare the approaches on the supplied BraTS20 Training 001 MRI volume and select a practical segmentation pipeline for subsequent project development. The three papers use closely related intensity-based ideas, but differ in the sophistication of their thresholding and post-processing stages.

2 Objective

The objectives for this week were:

- Study the segmentation methodology presented in all three papers.
- Implement the main processing stages described by each paper.
- Run the three approaches on the supplied BraTS20 FLAIR MRI data.
- Compare the resulting masks with the provided ground-truth segmentation using Dice similarity and Jaccard index.
- Select one approach for continued project development based on the observed segmentation quality, methodological fit, and implementation practicality.

3 Literature Review

3.1 Paper 1 – Hybridized Multilevel Thresholding and Level Set

Dawngliana et al. propose a hybrid method in which MRI images are first pre-processed using median filtering and histogram equalization, followed by multilevel thresholding based on Type-II fuzzy sets. Morphological processing removes small noisy structures, and a level-set method is then used to refine the tumor boundary. The paper reports that its MLT–Levelset method achieved an average Dice similarity of 1.48/2 and a Jaccard index of 0.74 on its evaluated cases, compared with 1.05/2 and 0.53 for the referenced FCM–Levelset method. [1]

3.2 Paper 2 – Thresholding and Morphological Operations

Murthy and Sadashivappa use a comparatively simple pipeline: MRI acquisition, Sobel-based pre-processing, histogram equalization, border removal, thresholding, and binary morphological operations such as erosion, dilation, opening and closing. The high-intensity cluster is treated as the tumor region, after which tumor features such as area, perimeter and centroid are extracted. The authors explicitly emphasize that the method is simple, computationally fast and easier to implement than more complex hybrid segmentation approaches. [2]

3.3 Paper 3 – Revised Integrated Thresholding and Morphological Process

Islam, Imteaz and Marium-E-Jannat use FLAIR images from the BRATS database. Their revised pipeline performs grayscale conversion, filtering/enhancement, global thresholding/Otsu thresholding, restoration of missing nearby tumor pixels, morphological erosion and dilation, removal of components smaller than 250 pixels, hole filling, and tumor-region analysis. The paper reports successful detection of 66 out of 76 FLAIR cases, corresponding to 86.84%. It also notes that performance is better for relatively clear and less noisy images. [3]

Table 1: Comparison of the three investigated segmentation pipelines.

Paper	Main segmentation idea	Main refinement	Reported/observed focus
Paper 1	Multilevel thresholding + level set	Morphology + bound-ment and higher	Boundary refineary evolution segmentation similarity
Paper 2	Global thresholding + morphology	Erosion/dilation/opening/closing	Simple, fast and low-complexity implementation
Paper 3	Otsu/global thresholding + restoration	Erosion/dilation + hole filtering and area analysis	Detection, location filling + size

4 Work Completed This Week

4.1 Methodology

A common experimental setup was used for the three approaches. The supplied BraTS20_Training_001 volume was unpacked and the FLAIR modality and segmentation ground truth were used. The axial slice containing the largest number of ground-truth tumor pixels was selected for the first implementation comparison (slice 67, using zero-based volume indexing).

For Paper 1, the implementation followed the paper's processing sequence: median filtering, histogram equalization, multilevel intensity partitioning, morphological cleanup and level-set refinement. The Type-II fuzzy/Differential Evolution optimization described mathematically in the paper was approximated for this first prototype by multilevel histogram thresholding, while the boundary-refinement stage was implemented with an equivalent image-based level-set/active-contour formulation. This distinction is important: the experiment is a practical reproduction of the pipeline, not a claim of exact reproduction of every optimization equation in the original implementation. [1]

For Paper 2, the implementation followed the stated sequence of filtering/enhancement, thresholding and morphological processing. The resulting binary components were filtered to remove small objects and obtain the tumor-region mask. [2]

For Paper 3, the implementation followed the revised integrated concept: FLAIR input, filtering/enhancement, global/Otsu thresholding, restoration of nearby pixels, morphological erosion/dilation, hole filling and removal of components below the paper's 250-pixel criterion. [3]

4.2 Implementation Details

The prototype was implemented in Python using NumPy, SciPy and scikit-image for image filtering, histogram processing, morphology, connected-component analysis and level-set refinement. The supplied ground truth

was used only for evaluation, not as an input to the segmentation algorithms.

The following metrics were calculated:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}, \quad \text{Jaccard} = \frac{|P \cap G|}{|P \cup G|}, \quad (1)$$

where P is the predicted tumor mask and G is the ground-truth mask. These metrics are consistent with the evaluation framework discussed in Paper 1. [1]

5 Challenges Faced

- The three papers describe different data formats and experimental settings, so their pipelines cannot be reproduced as identical code without adaptation.
- Paper 1 uses Type-II fuzzy membership functions and Differential Evolution for multilevel threshold optimization, which requires more detailed mathematical implementation than the other two methods.
- The third paper contains several empirically selected operations and parameters, including restoration of nearby pixels and a 250-pixel component criterion; these can be sensitive to image intensity and scale.
- A single 2D slice is not sufficient to establish overall segmentation performance for a 3D brain-tumor dataset.

6 Plan for Next Week

- Test the selected Paper 1 pipeline over multiple tumor-containing slices and additional BraTS cases.
- Improve the multilevel thresholding stage toward the Type-II fuzzy/Differential Evolution formulation described in the paper.
- Tune the level-set initialization and morphological parameters using validation data rather than a single slice.
- Add automated tumor localization and feature extraction such as area, centroid and bounding box.
- Record Dice, Jaccard, precision, recall and processing time across the validation cases.
- Prepare a reproducible OpenCV/Python implementation and organize the project repository.

References

- [1] M. Dawngliana, D. Deb, M. Handique and S. Roy, "Automatic Brain Tumor Segmentation in MRI: Hybridized Multilevel Thresholding and Level Set," *2015 International Symposium on Advanced Computing and Communication (ISACC)*, 2015.
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- [3] M. R. Islam, M. R. Imteaz and M.-E.-Jannat, "Detection and analysis of brain tumor from MRI by Integrated Thresholding and Morphological Process with Histogram based method," 2014.